Subject: Linux
Assignment: Linux Commands
Name: Khadra Hamud Abdullahi
ID: C6240063

CHAPTER 1: INTRODUCTION TO LINUX AND UBUNTU

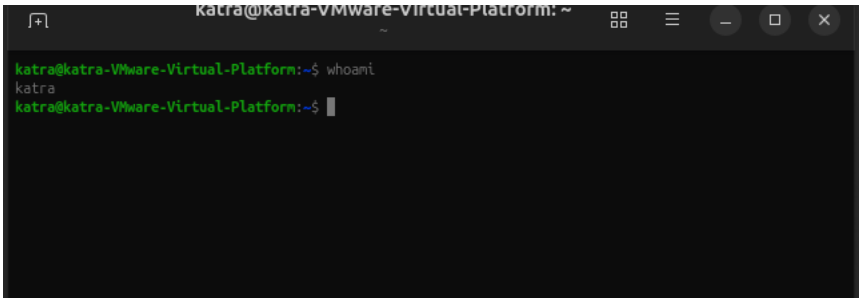
whoami Command

Definition:

The whoami command displays the username of the currently logged-in user. It helps users identify which account they are using in the Linux system.

Command Options:

The basic whoami command does not require any options in this chapter.



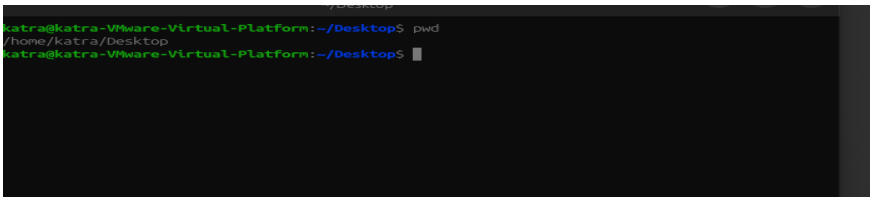
pwd Command (print working Directory)

Definition:

The pwd command stands for Print Working Directory. It displays the full path of the directory in which the user is currently working.

Command Options:

No options are introduced for this command in Chapter 1.



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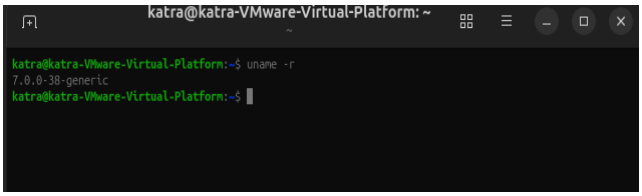
uname Command (unix name)

Definition:

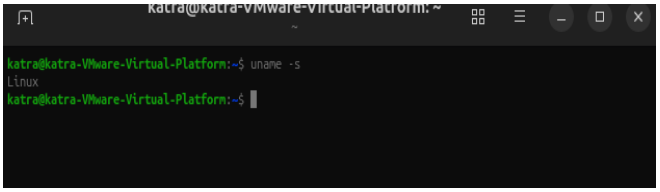
The uname command displays information about the operating system and the computer's kernel. It helps users identify the system they are working with.

Command Options:

-r – Displays the kernel release version.



-s – Displays the kernel name.



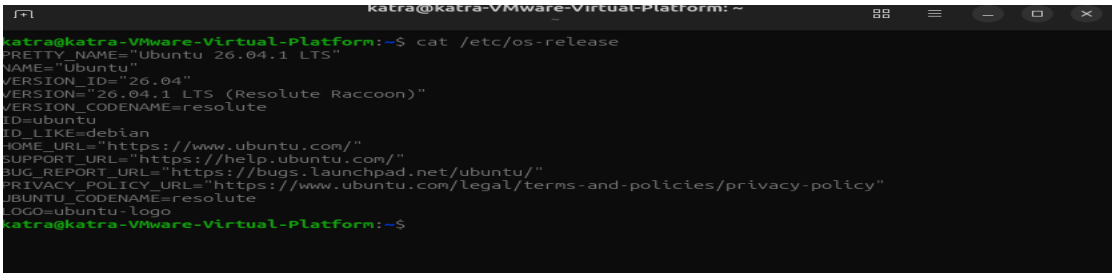
cat Command (concatenate)

Definition:

The cat command stands for Concatenate. It is used to display the contents of a file on the terminal. It can also combine the contents of multiple files.

Command Options:

No specific options are introduced for this command in Chapter 1.



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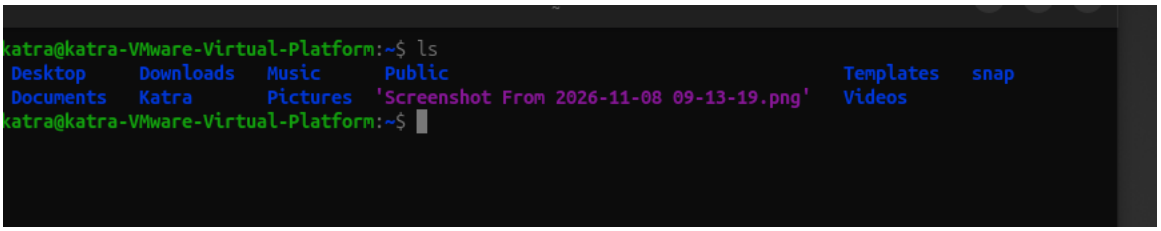
ls Command (List)

Definition:

The ls command stands for List. It displays the files and directories available in the current directory. It helps users view the contents of a directory.

Command Options:

No specific options are introduced for this command in Chapter 1.



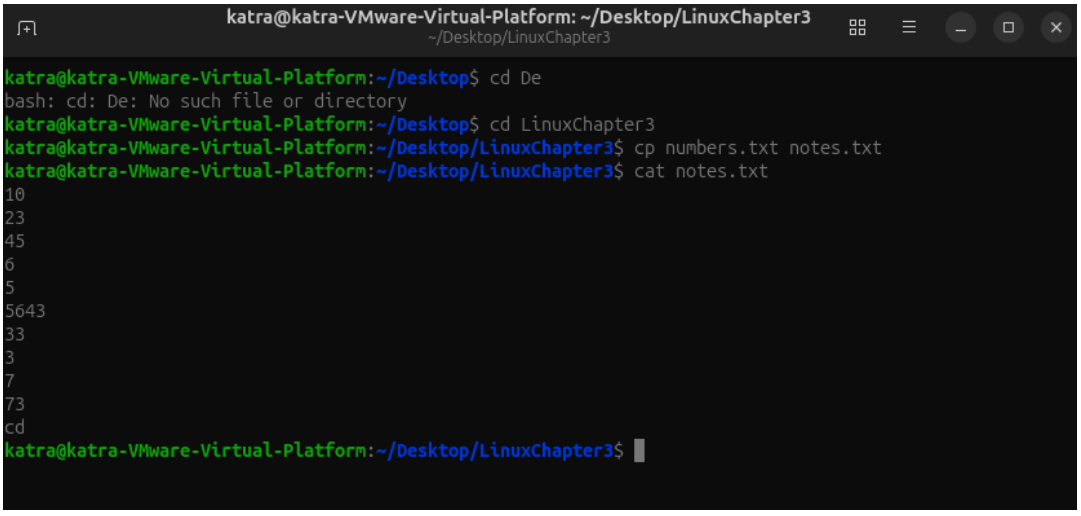
cp Command (Copy)

Definition:

The cp command stands for Copy. It is used to copy files or directories from one location to another without removing the original item.

Command Options:

No specific options are introduced for this command in Chapter 1.



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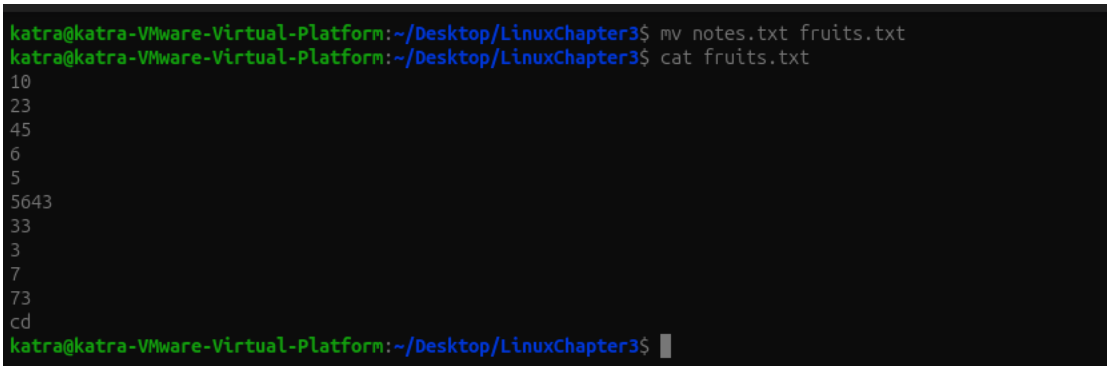
mv Command (Move)

Definition:

The mv command stands for Move. It is used to move files or directories from one location to another. It can also rename files and directories.

Command Options:

No specific options are introduced for this command in Chapter 1.



apt Command (Advanced package tool)

Definition:

The apt command stands for Advanced Package Tool. It is used in Ubuntu and other Debian-based Linux distributions to manage software packages. It allows users to install, update, upgrade, and remove software.

Command Options and Common Commands:

update – Refreshes the list of available software packages.

upgrade – Installs available updates for installed packages.

install – Installs a software package.

remove – Removes an installed package.

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dnf Command (Dandified YUM)

Definition:

The dnf command stands for Dandified YUM. It is a package management tool used in Fedora and some other RPM-based Linux distributions. It helps users install, update, and remove software packages.

Command Options and Common Commands:

install – Installs a software package.

update – Updates installed software packages.

remove – Removes an installed package.

search – Searches for available packages.

yum Command (Yellowdog Updater)

Definition:

The yum command stands for Yellowdog Updater, Modified. It is a package management tool used in older versions of Red Hat-based Linux distributions. It helps users install, update, and remove software packages.

Command Options and Common Commands:

install – Installs a software package.

update – Updates installed packages.

remove – Removes an installed package.

search – Searches for software package.

zypper Command

Definition:

The zypper command is a command-line package management tool used in SUSE Linux distributions. It allows users to install, update, search for, and remove software packages.

Command Options and Common Commands:

install – Installs a software package.

update – Updates installed packages.

remove – Removes an installed package.– Searches for available search packages.

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CHAPTER 2 – Linux System Architecture, Command Line, and File Management

pwd Command

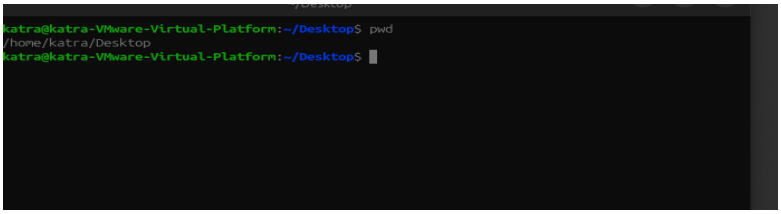
Abbreviation: Print Working Directory.

Definition:

The pwd command is used to display the full path of the current working directory.

Command Options and Common Commands:

- pwd – Displays the current working directory.



cd Command

Abbreviation: Change Directory.

Definition:

The cd command is used to change the current working directory and move between directories.

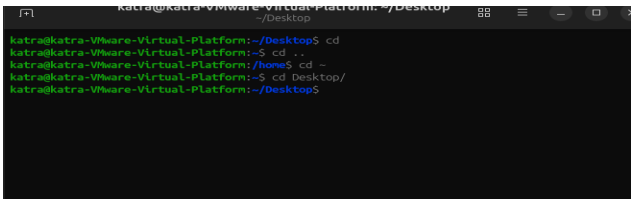
Command Options and Common Commands:

cd – Moves to the home directory.

cd .. – Moves one directory level up.

cd ~ – Moves to the home directory.

cd - – Returns to the previous directory.



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ls Command

Abbreviation**: List (Directory Contents).**

Definition:

The ls command is used to display files and directories in the current directory.

Command Options and Common Commands:

ls – Lists files and directories.

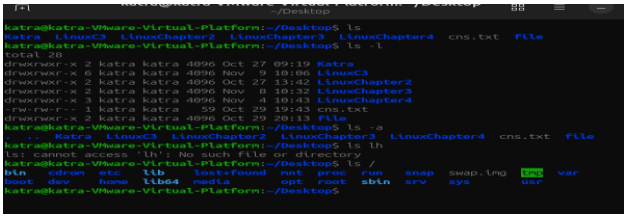
-a – Displays all files, including hidden files.

-l – Displays files in long format with details.

-h – Displays file sizes in human-reada

ble format.

-lah – Combines all three options.



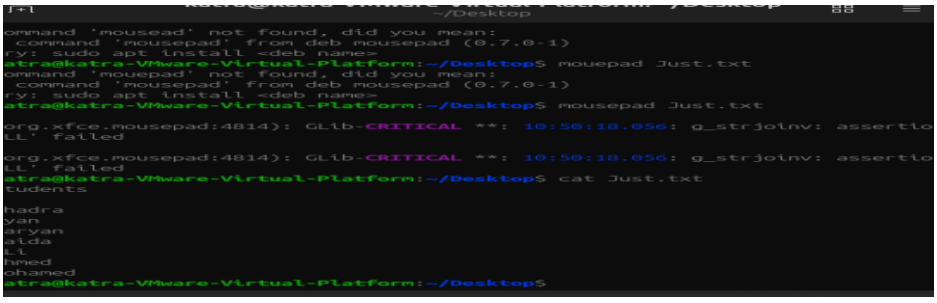
touch Command

Abbreviation: Not an abbreviation; it is a regular command name.

Definition:

The touch command is used to create empty files or update the timestamps of existing files.

touch file.txt – Creates an empty file.



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mkdir Command

Abbreviation: Make Directory.

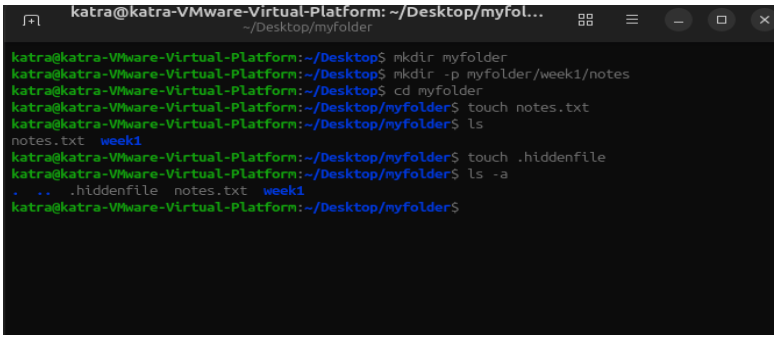
Definition:

The mkdir command is used to create new directories (folders) in Linux.

Command Options and Common Commands:

mkdir folder – Creates a new directory.

mkdir -p – Creates parent directories when needed.



rmdir Command

Abbreviation: Remove Directory.

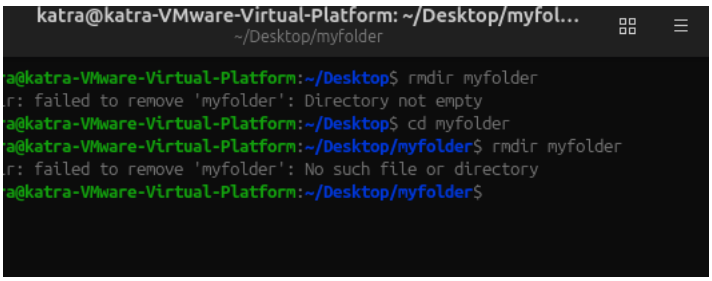
Definition:

The rmdir command is used to remove empty directories in Linux.

Command Options and Common Commands:

rmdir folder – Removes an empty directory.

It cannot remove a directory containing files.



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Copy (cp) Command

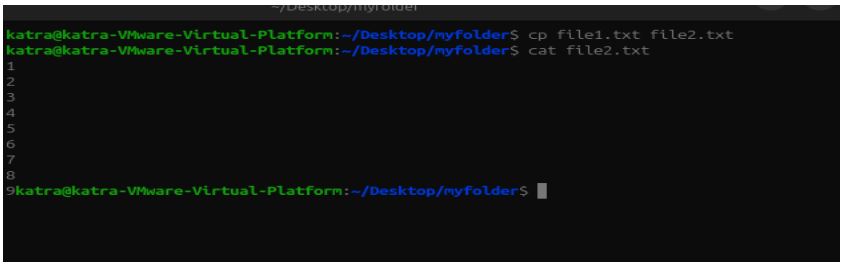
Definition:

The copy command is used to copy files and directories from one location to another.

Command Options and Common Commands:

cp – Copies a file.

cp -r – Copies directories and their contents recursively.



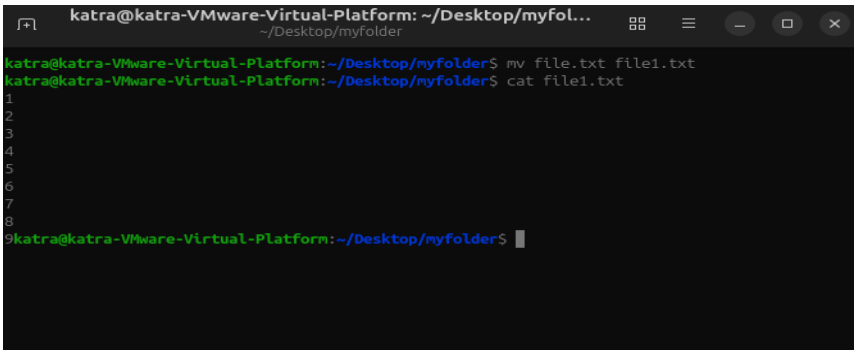
Move (mv) Command

Definition:

The move command is used to move files and directories from one location to another or rename them.

Command Options and Common Commands:

mv – Moves a file or directory.



Remove (rm) Command

Definition:

The remove command is used to delete files and directories.

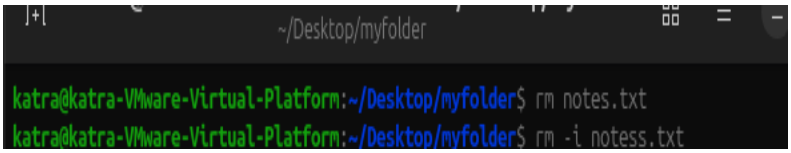
Subject: Linux
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ID: C6240063

Command Options and Common Commands:

rm – Removes a file.

rm -i – Asks for confirmation before deleting.

rm -r – Removes directories and their contents recursively.



Manual (man) Command

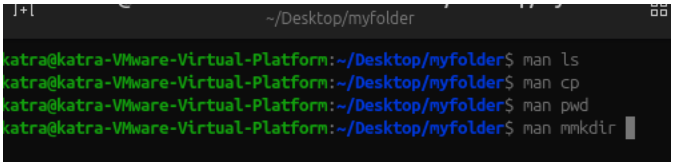
Definition:

The man command is used to display the manual pages of Linux commands.

Command Options and Common Commands:

man command – Displays the manual of a command.

man ls – Displays the manual for the ls command.



Info Command

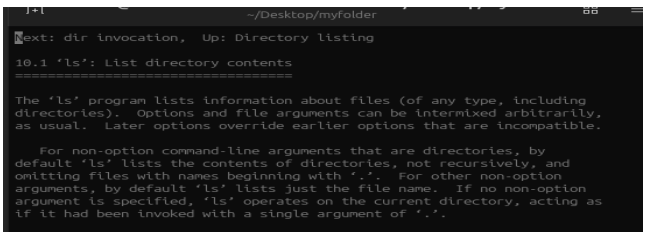
Definition:

The info command is used to display detailed information about Linux commands and programs.

Command Options and Common Commands:

info – Opens the information documentation.

info ls – Displays information about the ls command.



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Help Command

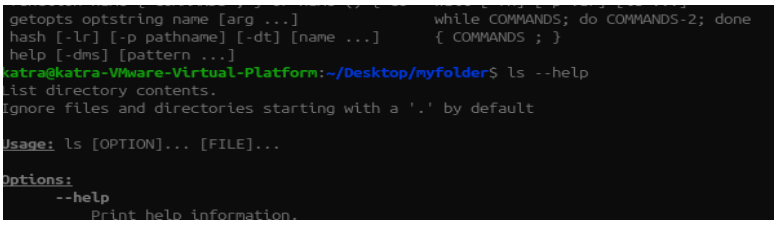
Definition:

The help command is used to display information about Bash built-in commands.

Command Options and Common Commands:

help – Displays information about built-in commands.

help cd – Displays help about the cd command.



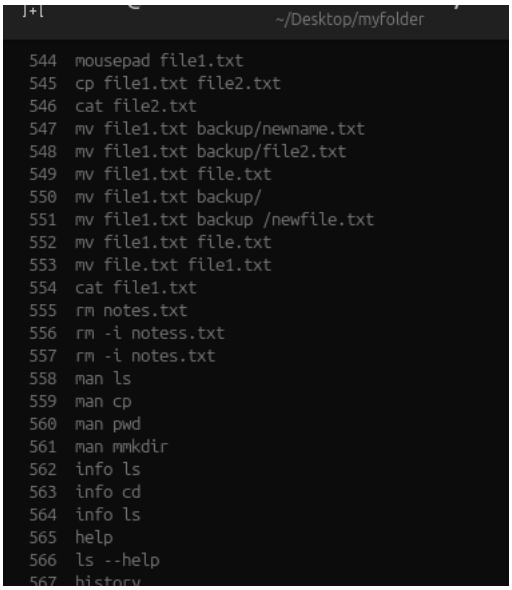
History Command

Definition:

The history command is used to display previously executed commands in the terminal.

Command Options and Common Commands:

history – Displays command history.



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Echo Command

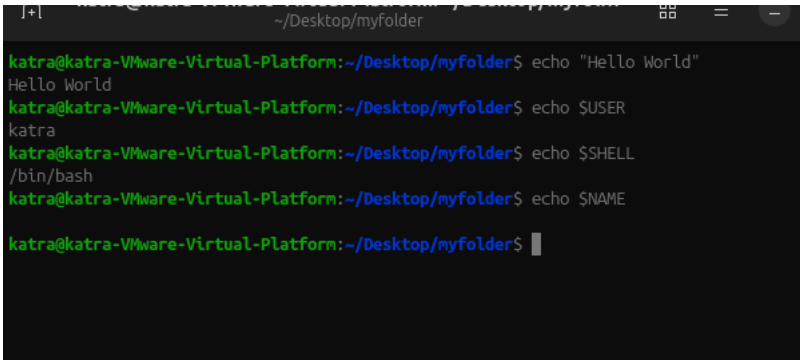
Definition:

The echo command is used to display text or variable values on the terminal.

Command Options and Common Commands:

echo Hello – Displays the word Hello.

echo $NAME – Displays the value of the NAME variable.



Export Command

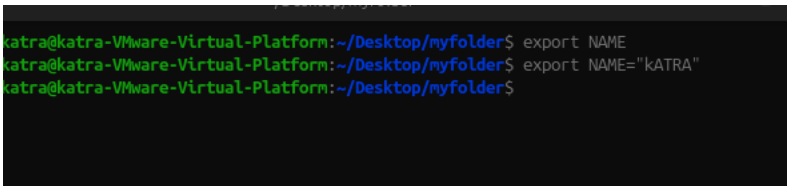
Definition:

The export command is used to make variables available to child processes in the shell environment.

Command Options and Common Commands:

export NAME=value – Exports a variable.

export PATH – Exports the PATH variable.



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CHAPTER 3 – Text Processing, Searching, Archiving, and Editors

CAT (CONCATENATE) COMMAND

Definition:

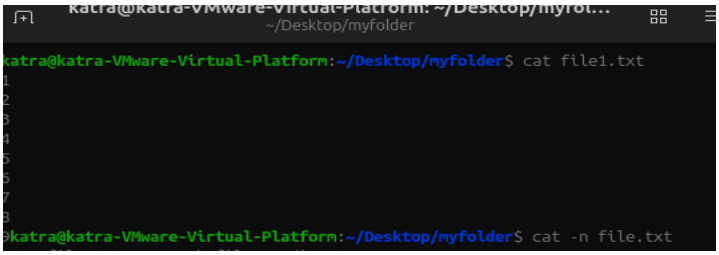
The cat command is used to display the contents of text files and combine multiple files into one.

Command Options and Common Commands:

cat – Displays file contents.

cat -n – Displays file contents with line numbers.

cat file1 file2 – Displays multiple files in sequence.



LESS COMMAND

Definition:

The less command is used to read long text files one screen at a time without modifying their contents.

Command Options and Common Commands:

less filename – Opens a file for reading.

Space – Moves forward one screen.

b – Moves backward one screen.

/word – Searches for a word.

n – Finds the next match.

q – Exits less.

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MORE COMMAND

Definition:

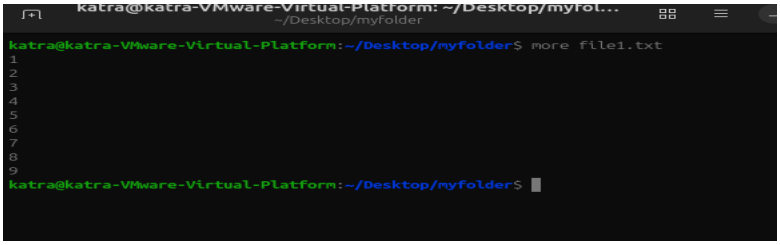
The more command is used to read long text files one screen at a time.

Command Options and Common Commands:

more filename – Opens a file for reading.

Space – Moves forward one screen.

q – Exits more.



HEAD COMMAND

Definition:

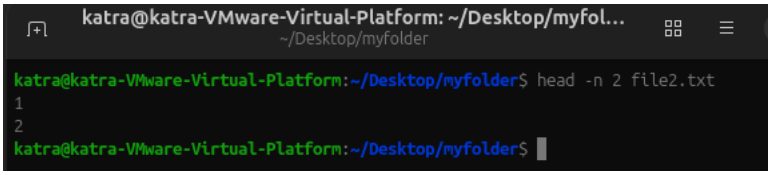
The head command is used to display the beginning of a file.

Command Options and Common Commands:

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head filename – Displays the first 10 lines by default.

head -n 2 filename – Displays the first 2 lines.



TAIL COMMAND

Definition:

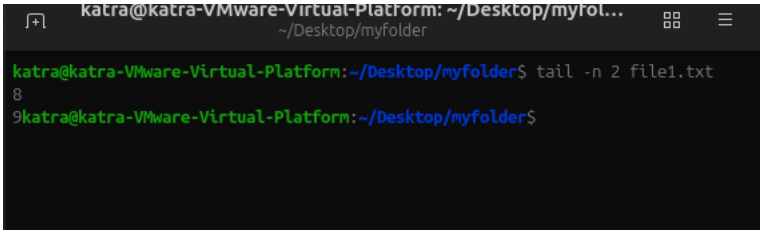
The tail command is used to display the end of a file.

Command Options and Common Commands:

tail filename – Displays the last 10 lines by default.

tail -n 2 filename – Displays the last 2 lines.

tail -f filename – Monitors new lines added to a file in real time.



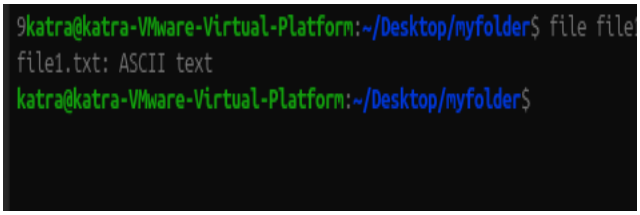
FILE COMMAND

Definition:

The file command is used to identify the actual type of a file.

Command Options and Common Commands:

file filename – Identifies the file type.



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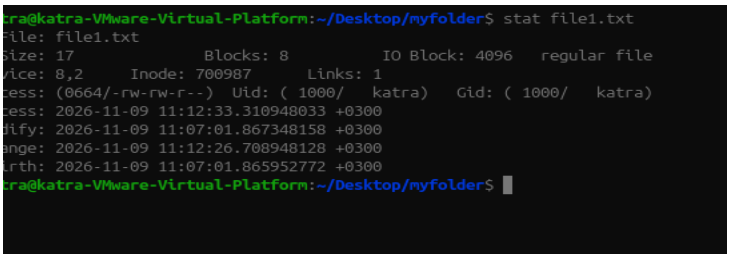
STAT COMMAND

Definition:

The stat command is used to display detailed file metadata.

Command Options and Common Commands:

stat filename – Displays file information such as size, permissions, owner, and timestamps.



WHICH COMMAND

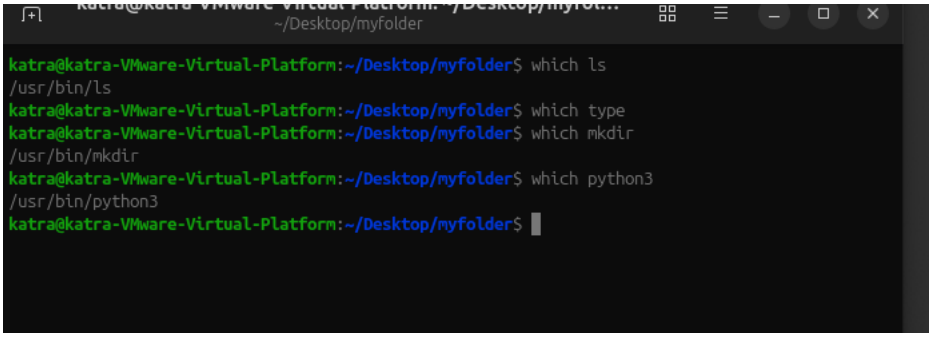
Definition:

The which command is used to locate executable programs by searching directories in the PATH environment variable.

Command Options and Common Commands:

which command – Finds the location of an executable command.

which ls – Finds the location of the ls executable.



TYPE COMMAND

Definition:

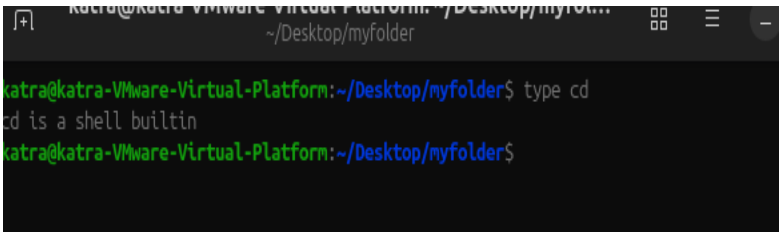
The type command is used to determine whether a command is a shell built-in, executable file, alias, or function.

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Command Options and Common Commands:

type command – Identifies the type of a command.

type cd – Checks whether cd is a shell built-in.



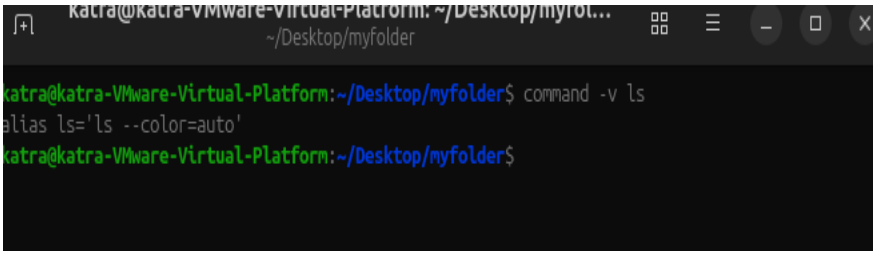
COMMAND -V

Definition:

The command -v command is used to find how a command is resolved by the shell.

Command Options and Common Commands:

command -v ls – Displays how the ls command is located.



WHEREIS COMMAND

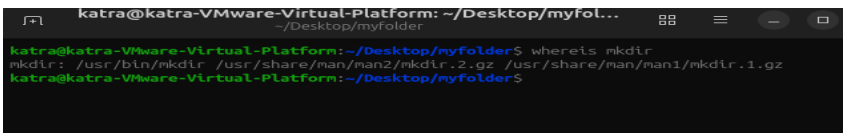
Definition:

The whereis command is used to locate the executable binary, source code, and manual pages associated with a Linux command.

Command Options and Common Commands:

whereis command – Locates files related to a command.

whereis tar – Locates the tar program and its manual pages.



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FIND COMMAND

Definition:

The find command is used to search for files and directories based on their names, types, sizes, permissions, and other conditions.

Command Options and Common Commands:

find . -name 'names.txt' – Searches for a file by name.

find . -name '*.log' – Searches for files ending in .log.

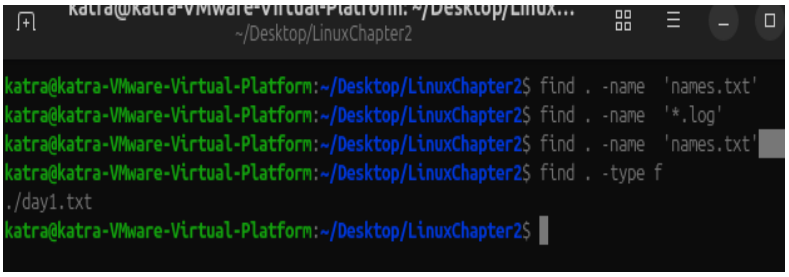
find project -type d – Searches for directories.

find . -type f – Searches for regular files.

find . -iname 'names.txt' – Searches without considering letter case.

find . -size +10c – Searches for files larger than 10 bytes.

find . -mmin -30 – Searches for files modified within the last 30 minutes.



LOCATE COMMAND

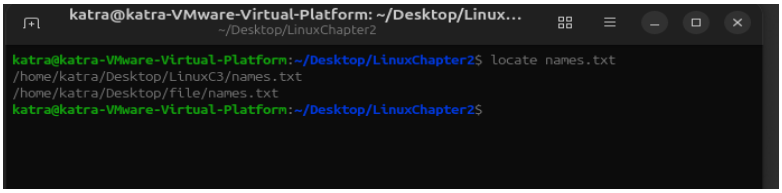
Definition:

The locate command is used to find files and directories quickly by searching a prebuilt database of file paths.

Command Options and Common Commands:

locate filename – Searches for a file using the database.

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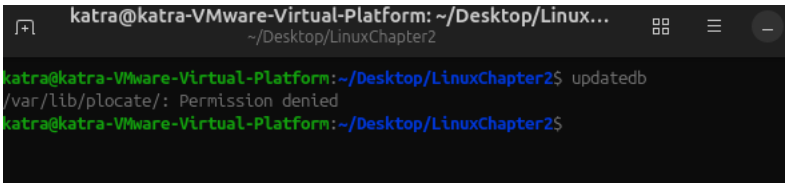
UPDATEDB COMMAND

Definition:

The updatedb command is used to update the database that the locate command searches.

Command Options and Common Commands:

updatedb – Refreshes the file location database.



GREP (GLOBAL REGULAR EXPRESSION PRINT) COMMAND

Definition:

The grep command is used to search for specific text or patterns inside files. It displays only the lines that match the search pattern.

Command Options and Common Commands:

grep 'Ali' names.txt – Searches for Ali inside names.txt.

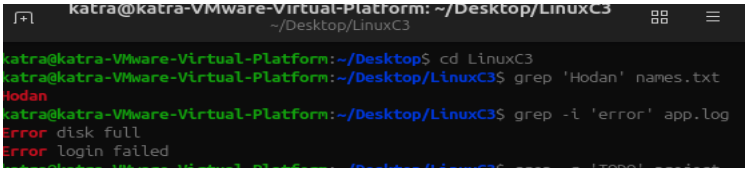
grep -n 'ERROR' app.log – Displays matching lines with line numbers.

grep -i 'error' app.log – Searches without considering letter case.

grep -v 'ERROR' app.log – Displays lines that do not match ERROR.

grep -r 'TODO' project – Searches recursively inside directories.

grep -rn 'TODO' project – Searches recursively and displays line numbers.



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REDIRECTION OPERATORS

Definition:

Redirection is used to change where a command receives input or sends output.

Common Operators:

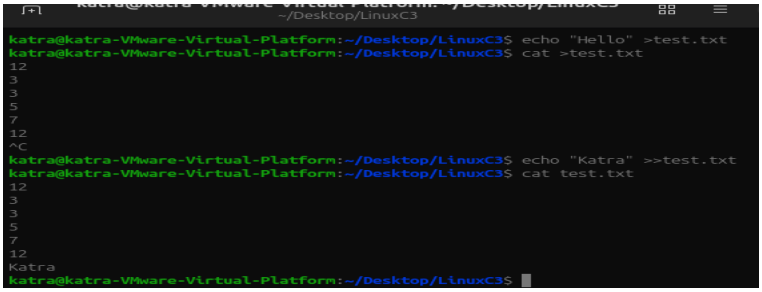
> – Sends output to a file and overwrites existing content.

>> – Appends output to the end of a file.

2> – Redirects error messages to a file.

< – Reads input from a file.

| – Sends the output of one command to another command.



SORT COMMAND

Definition:

The sort command is used to arrange lines of text alphabetically by default.

Command Options and Common Commands:

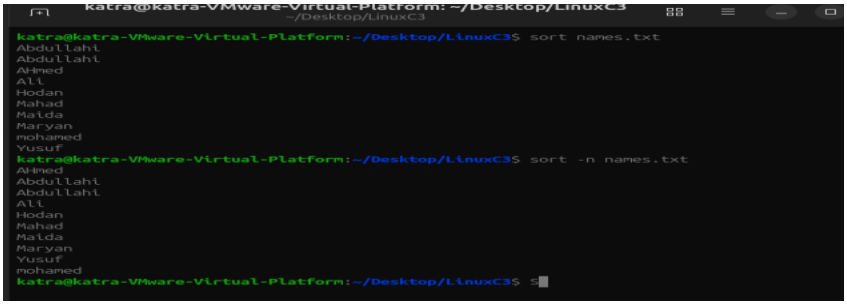
sort filename – Sorts lines alphabetically.

sort -n filename – Sorts numbers numerically.

sort -r filename – Reverses the sorting order.

sort -rn filename – Sorts numbers in reverse numerical order.

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UNIQ COMMAND

Definition:

The uniq command is used to remove repeated adjacent lines from text.

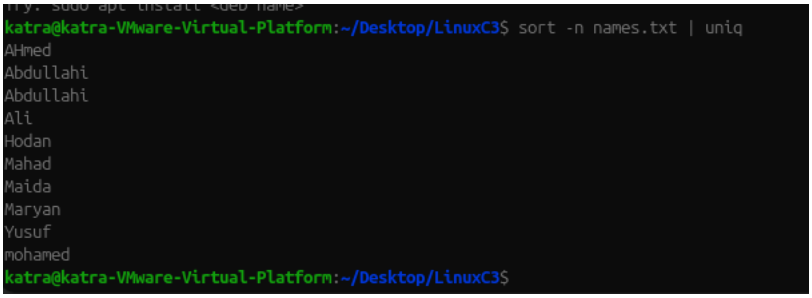
Command Options and Common Commands:

uniq – Removes adjacent duplicate lines.

uniq -c – Counts repeated lines.

uniq -d – Displays duplicate lines.

sort filename | uniq – Sorts and removes duplicate lines.



CUT COMMAND

Definition:

The cut command is used to extract specific parts of text from a file or command output.

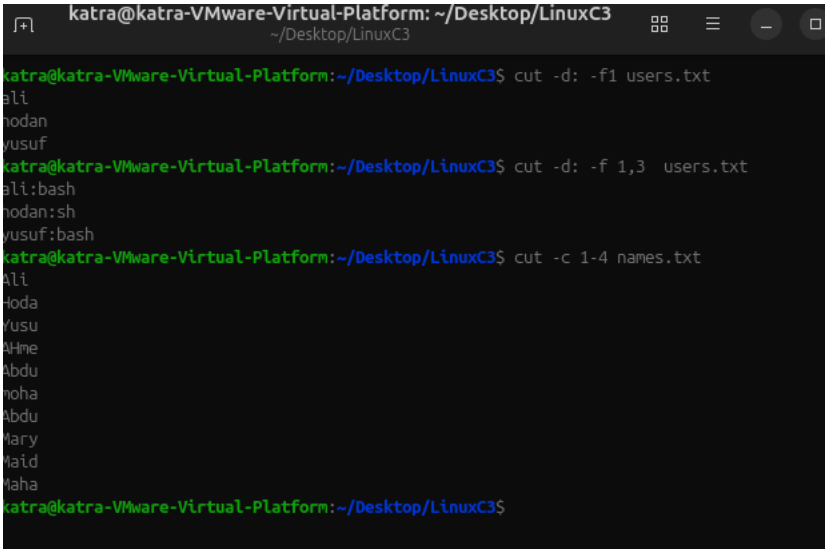
Command Options and Common Commands:

cut -d: -f1 users.txt – Extracts the first field using a colon delimiter.

cut -d: -f 1,3 users.txt – Extracts fields 1 and 3.

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cut -c 1-4 filename – Extracts the first four characters of each line.



PASTE COMMAND

Definition:

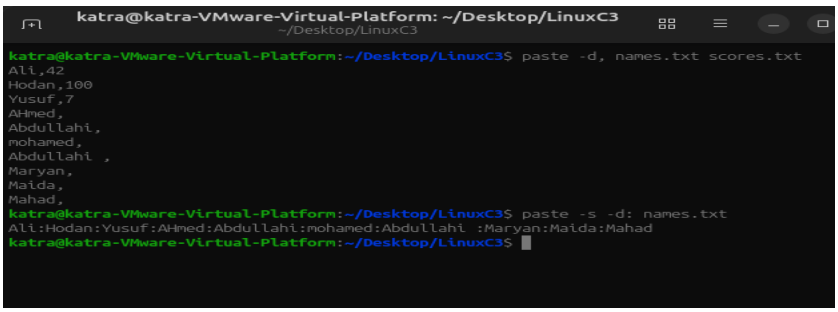
The paste command is used to combine corresponding lines from two or more files into a single line.

Command Options and Common Commands:

paste names.txt scores.txt – Combines corresponding lines using a tab.

paste -d, names.txt scores.txt – Uses a comma as the separator.

paste -s -d, names.txt – Combines all lines into one line using commas.



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WC (WORD COUNT) COMMAND

Definition:

The wc command is used to count the number of lines, words, and bytes in a file or command output.

Command Options and Common Commands:

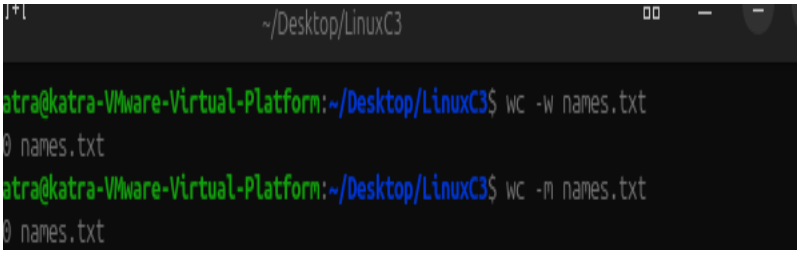
wc filename – Counts lines, words, and bytes.

wc -l filename – Counts lines.

wc -w filename – Counts words.

wc -c filename – Counts bytes.

wc -m filename – Counts characters.



TR (TRANSLATE) COMMAND

Definition:

The tr command is used to replace, delete, or remove repeated characters from text.

Command Options and Common Commands:

tr 'a-z' 'A-Z' – Converts lowercase letters to uppercase.

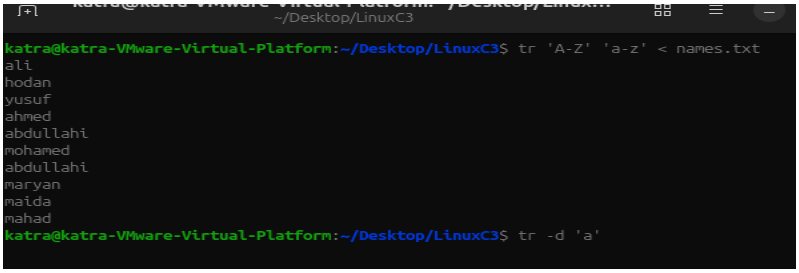
tr 'A-Z' 'a-z' – Converts uppercase letters to lowercase.

tr -d 'a' – Deletes the selected character.

tr -s ' ' – Removes repeated spaces.

tr 'a-z' 'A-Z' < names.txt – Converts text in a file to uppercase.

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TAR (TAPE ARCHIVE) COMMAND

Definition:

The tar command is used to combine multiple files or directories into one archive file. It can also work with compression tools.

Command Options and Common Commands:

tar -cf archive.tar filename – Creates a new archive.

tar -cvf archive.tar directory/ – Creates an archive and displays filenames.

tar -xf archive.tar – Extracts files from an archive.

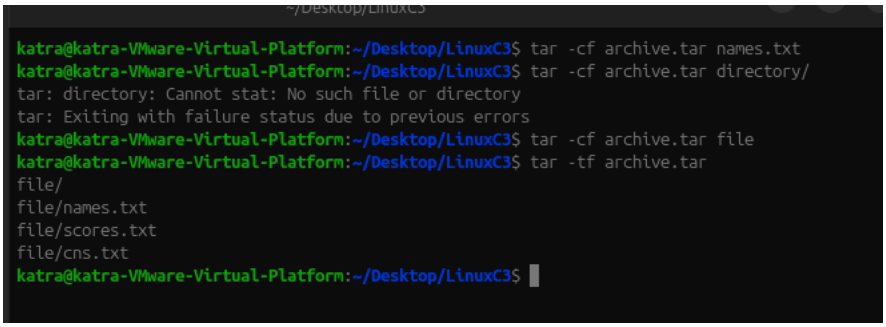
tar -tf archive.tar – Lists archive contents.

tar -czf archive.tar.gz directory/ – Creates a gzip-compressed archive.

tar -cjf archive.tar.bz2 directory/ – Creates a bzip2-compressed archive.

tar -cJf archive.tar.xz directory/ – Creates an xz-compressed archive.

tar -xf archive.tar -C directory/ – Extracts files into a specified directory.



TAR OPTIONS:

-c – Creates a new archive.

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-x – Extracts files from an archive.

-v – Displays processed filenames.

-f – Specifies the archive filename.

-t – Lists archive contents.

-z – Uses gzip compression.

-j – Uses bzip2 compression.

-J – Uses xz compression.

-C – Changes to a specified directory.

GZIP COMMAND

Definition:

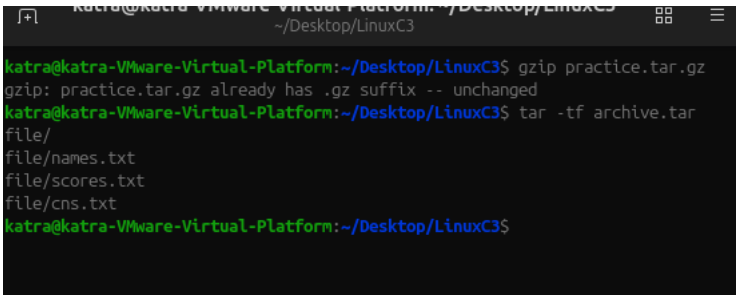
The gzip command is used to compress files to reduce their size.

Command Options and Common Commands:

gzip filename – Compresses a file.

gzip -k filename – Compresses a file while keeping the original.

gzip -9 filename – Uses the highest compression level.



GUNZIP COMMAND

Definition:

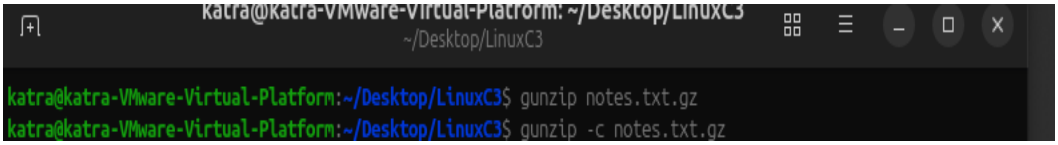
The gunzip command is used to decompress a .gz file and restore the original file.

Command Options and Common Commands:

gunzip filename.gz – Decompresses a .gz file.

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gunzip -c filename.gz – Displays the restored contents without removing the compressed file.



BZIP2 COMMAND

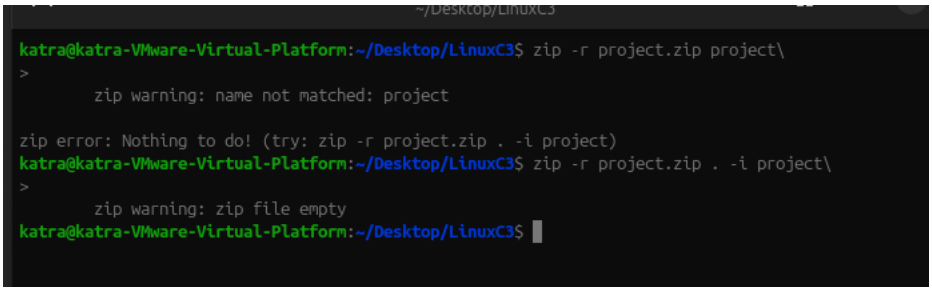
Definition:

The bzip2 command is used to compress files to reduce their size.

Command Options and Common Commands:

bzip2 filename – Compresses a file.

bzip2 -k filename – Compresses a file while keeping the original.



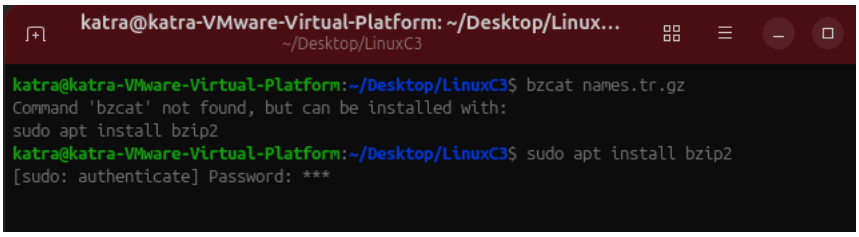
BZCAT COMMAND

Definition:

The bzcat command is used to display the contents of a bzip2-compressed file without extracting it.

Command Options and Common Commands:

bzcat filename.bz2 – Displays the contents of a compressed file.



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BUNZIP2 COMMAND

Definition:

The bunzip2 command is used to decompress bzip2-compressed files.

Command Options and Common Commands:

bunzip2 filename.bz2 – Extracts a .bz2 file.

XZ COMMAND

Definition:

The xz command is used to compress files and reduce their size.

Command Options and Common Commands:

xz filename – Compresses a file.

xz -k filename – Compresses a file while keeping the original.

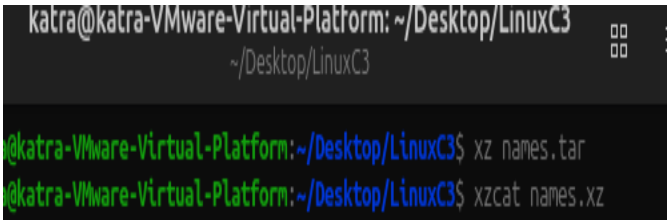
XZCAT COMMAND

Definition:

The xzcat command is used to display the contents of an xz-compressed file without extracting it.

Command Options and Common Commands:

xzcat filename.xz – Displays the contents of a compressed file.



UNXZ COMMAND

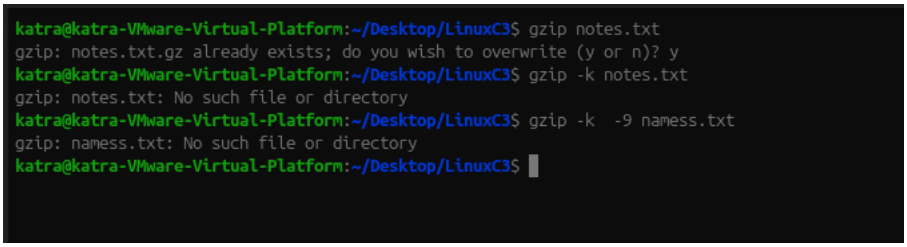
Definition:

The unxz command is used to decompress xz-compressed files.

Command Options and Common Commands:

unxz filename.xz – Extracts an .xz file.

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ZIP COMMAND

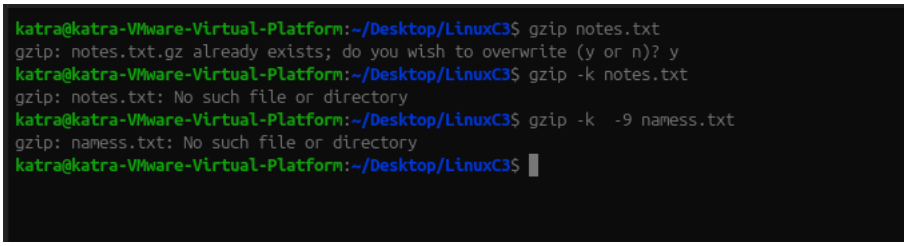
Definition:

The zip command is used to combine multiple files into a single archive and compress them to reduce storage space.

Command Options and Common Commands:

zip archive.zip filename – Creates a ZIP archive.

zip -r project.zip project/ – Compresses a directory and its contents recursively.



UNZIP COMMAND

Definition:

The unzip command is used to extract files from a ZIP archive.

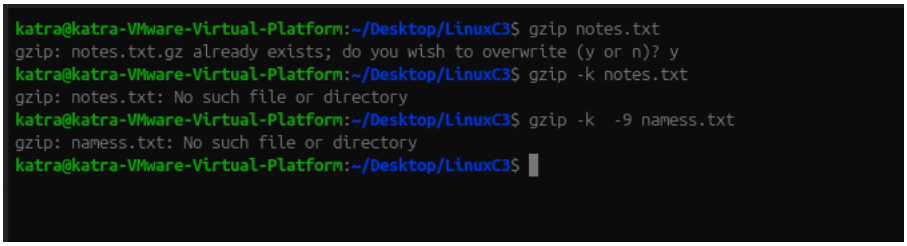
Command Options and Common Commands:

unzip archive.zip – Extracts a ZIP archive.

unzip -l archive.zip – Lists archive contents without extracting.

unzip archive.zip -d directory – Extracts files into a specified directory.

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NANO COMMAND

Definition:

The nano command is a beginner-friendly text editor used to create and edit text files directly in the terminal.

Command Options and Common Commands:

nano filename – Opens or creates a text file.

NANO SHORTCUTS:

Ctrl + O – Saves the file (Write Out).

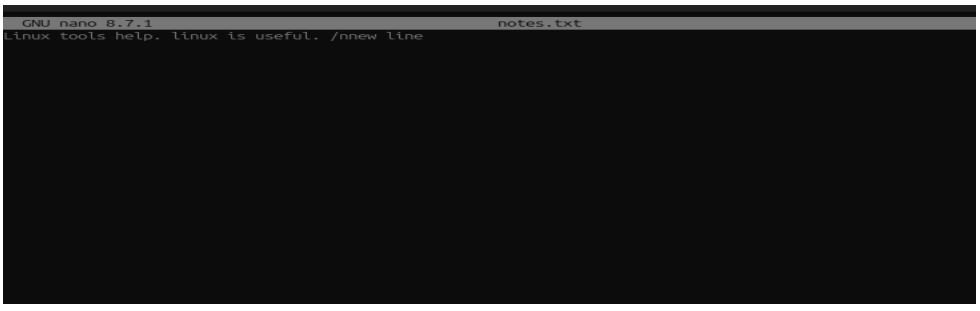
Ctrl + X – Exits Nano.

Ctrl + W – Searches for text.

Ctrl + K – Cuts the current line.

Ctrl + U – Pastes the cut line.

Alt + U – Undoes the last change.



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VIM COMMAND

Definition:

The vim command is a text editor used to create and edit text files directly in the terminal.

Command Options and Common Commands:

vim filename – Opens a file in Vim.

i – Enters Insert mode.

Esc – Returns to Normal mode.

:w – Saves the file.

:q – Exits Vim.

:wq – Saves and exits Vim.

:q! – Exits without saving changes.

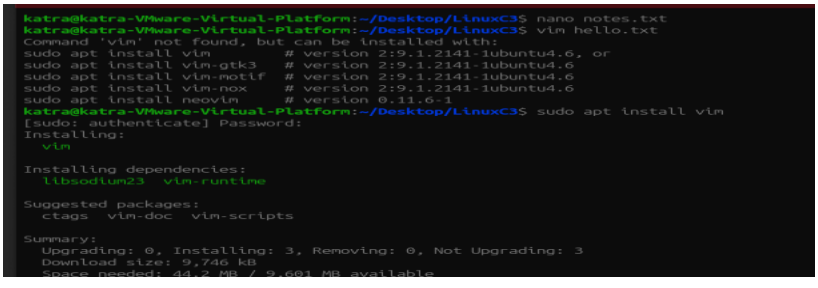
VI COMMAND

Definition:

The vi command is used to open a text editor in the terminal. On Ubuntu, vi opens the same editor as Vim.

Command Options and Common Commands:

vi filename – Opens a file in the editor.



VIMTUTOR COMMAND

Definition:

The vimtutor command is used to open a built-in tutorial that teaches users how to use Vim.

Command Options and Common Commands:

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vimtutor – Starts the Vim tutorial.

LN (LINK) COMMAND

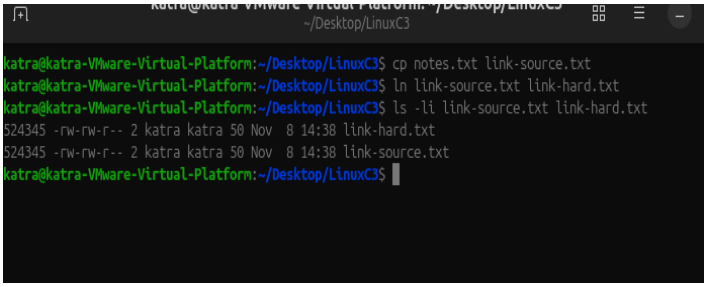
Definition:

The ln command is used to create another name for an existing file. By default, it creates a hard link.

Command Options and Common Commands:

ln source-file hard-link – Creates a hard link.

ls -li filename – Displays file information, including the inode number.



LINK (LN -S) COMMAND

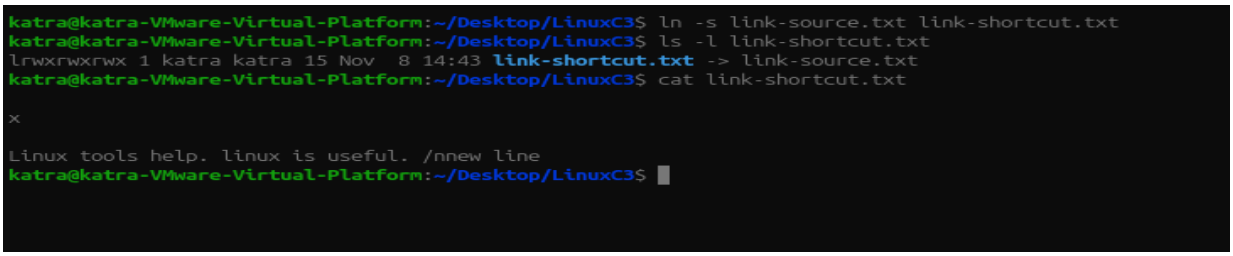
Definition:

A symbolic link is a special file that points to another file or directory using its path.

Command Options and Common Commands:

ln -s target-file symbolic-link – Creates a symbolic link.

ls -l symbolic-link – Displays the symbolic link and its target path.



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LS -LI COMMAND

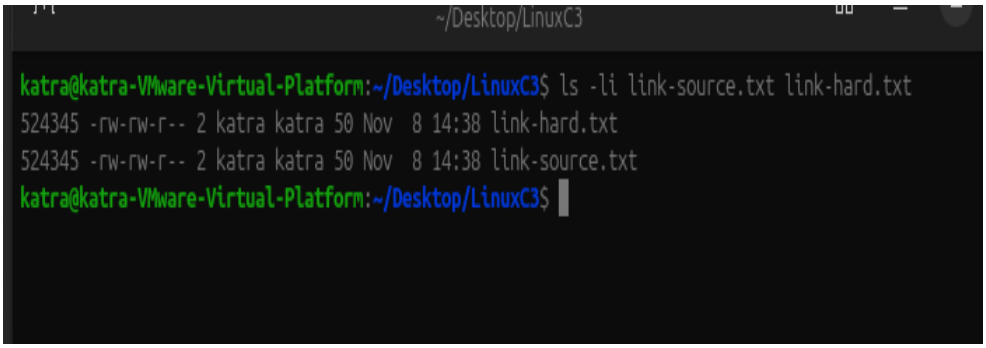
Definition:

The ls -li command is used to list files with detailed information, including their inode numbers.

Command Options and Common Commands:

ls -li – Displays files with inode numbers.

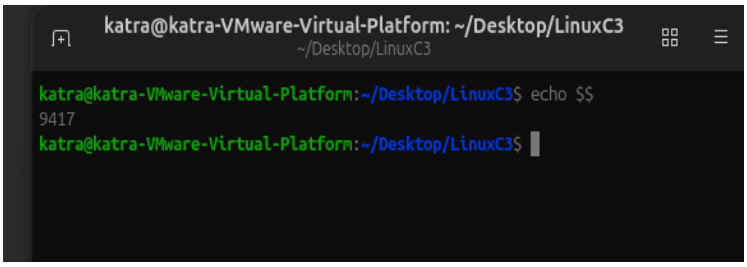
ls -li filename – Displays the inode number of a specific file.



CHAPTER 4: PROCESS, SERVICE, AND SYSTEM MANAGEMENT

1. PROCESSES AND JOBS

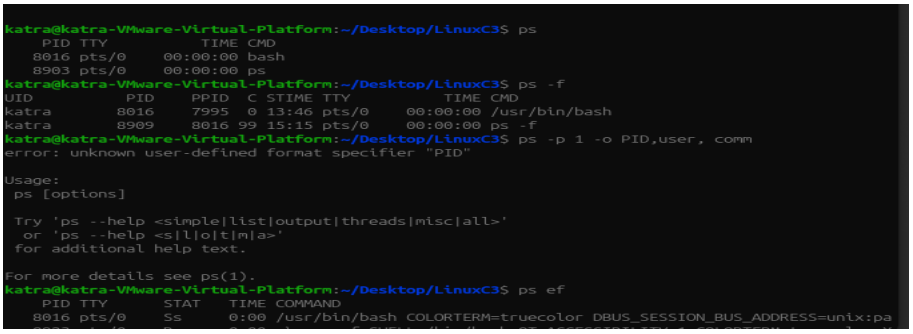
1. echo – Displays text or variable values. $$ – Displays the PID of the current shell.



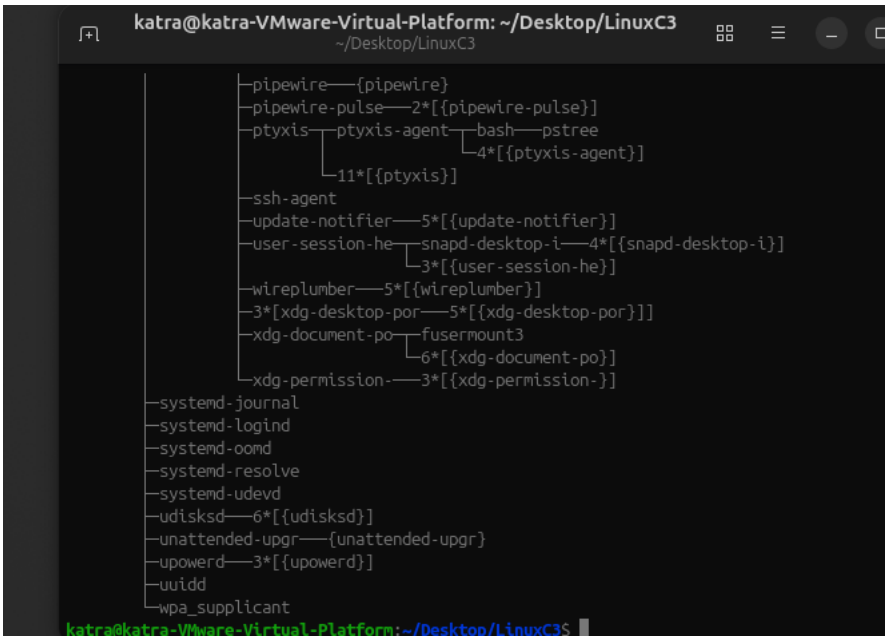
2. ps – Displays running processes. -f – Full format; -p PID – Displays a specific process; -o COLS – Selects displayed columns; aux – Displays all processes with CPU and memory information; -ef – Displays all

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processes in full format; --sort=-%mem – Sorts processes by memory usage, highest first.

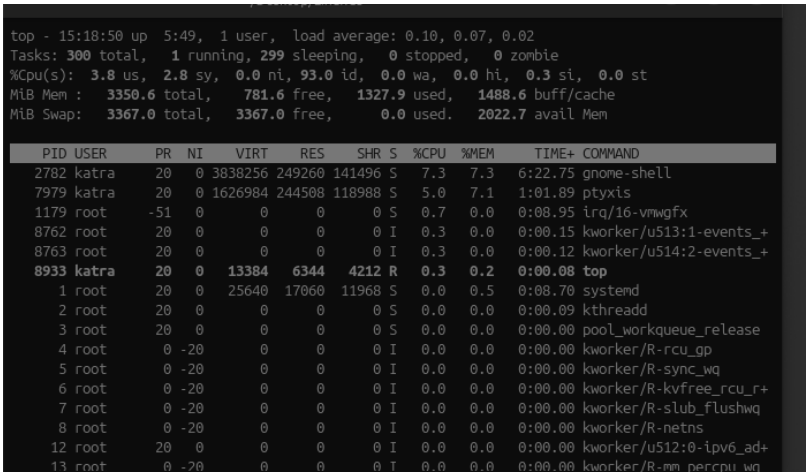


3. pstree – Displays processes in a tree structure. -p – Displays process IDs beside process names.

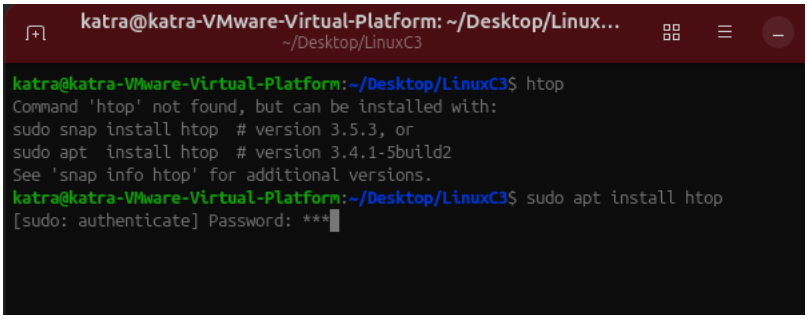


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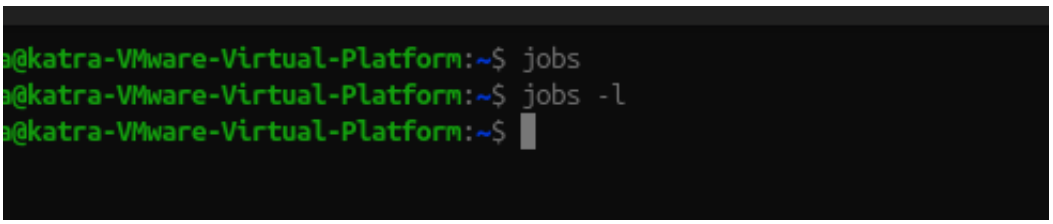
4. top – Displays running processes in real time. -u USER – Displays processes belonging to a user; -p PID – Displays a specific process; P – Sorts by CPU usage; M – Sorts by memory usage; k – Sends a signal to a process; r – Changes process priority; q – Quits.



5. htop – Displays processes in an interactive interface. -u USER – Displays a user's processes; -t – Displays processes in tree view; F6 – Selects sorting column; F9 – Sends a signal; F10 – Quits; q – Quits.



6. jobs – Displays background and stopped jobs. -l – Displays job numbers and PIDs; %N – Refers to a specific job number.

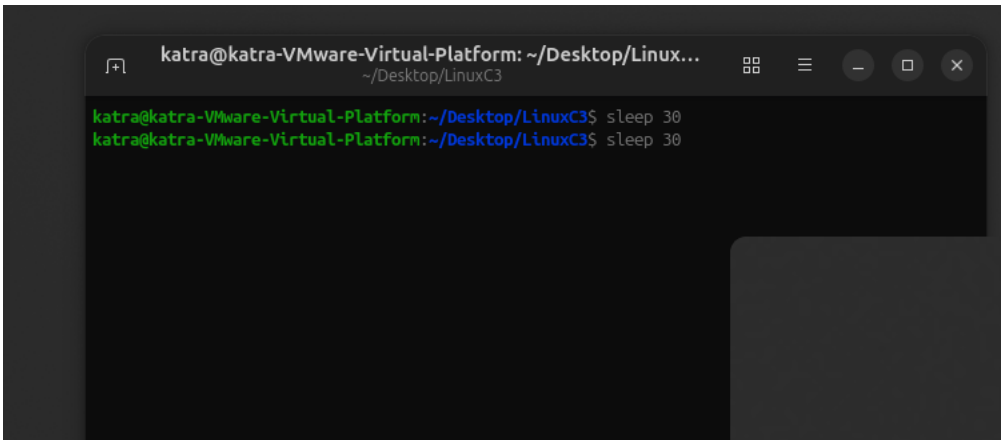


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7. bg – Resumes a stopped job in the background. %N – Specifies the job number.

8. fg – Brings a background or stopped job to the foreground. %N – Specifies the job number.

9. sleep – Delays execution for a specified time.



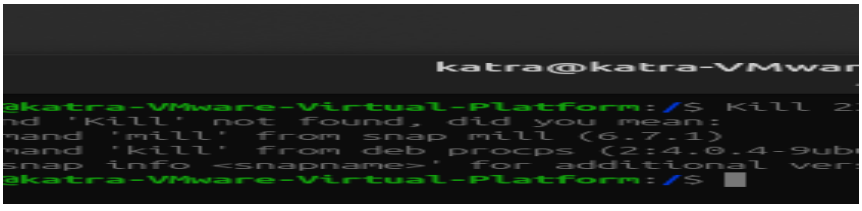
10. & – Runs a command in the background.

11. Ctrl+C – Sends SIGINT to interrupt a foreground process

12. Ctrl+Z – Sends SIGTSTP to pause a foreground process.

2. SIGNALS, KILL, AND PRIORITY

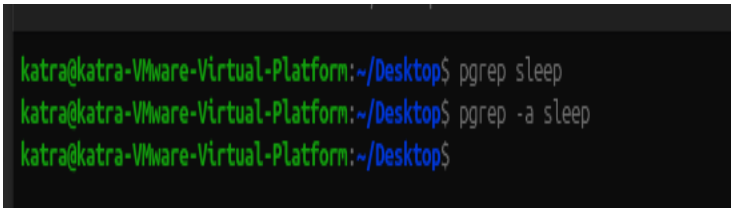
13. kill – Sends a signal to a process. -TERM – Requests process termination; -HUP – Sends a hang-up signal; -9 – Forcefully terminates a process using SIGKILL; %N – Specifies a job number.



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14. kill -l – Lists available signals. -l 15 – Displays the name of signal number 15.

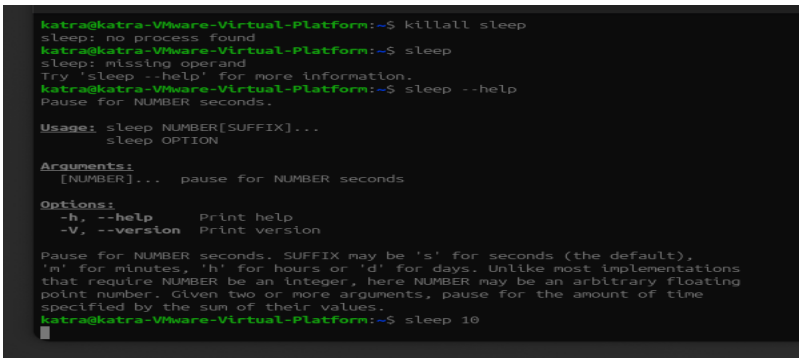
15. pgrep – Searches for processes by name. -a – Displays PID and full command line.



16. pkill – Sends a signal to processes matching a name or pattern. -u USER – Targets processes belonging to a specific user.



17. killall – Terminates processes by process name.

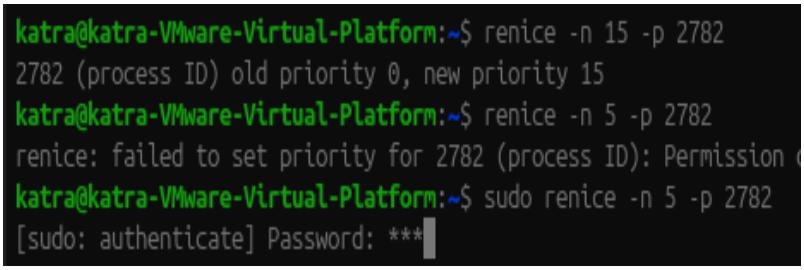


18. nice – Starts a process with a specified niceness value. -n N – Sets the niceness value. The range is -20 to 19; -20 means highest priority, 0 means default priority, and 19 means lowest priority.



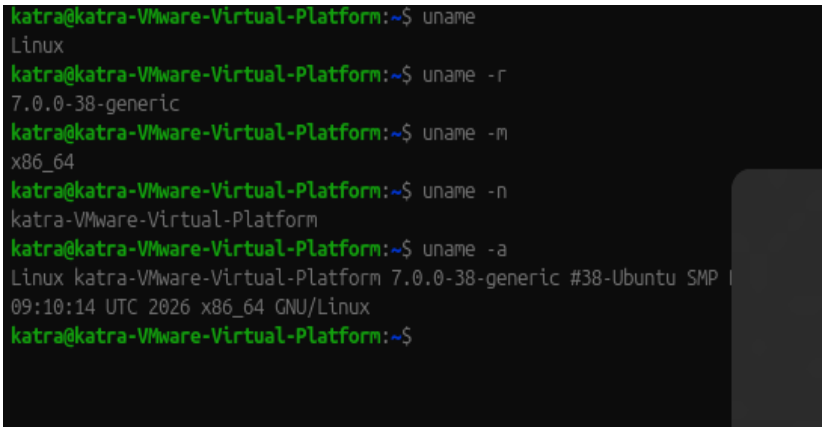
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19. renice – Changes the priority of an existing process. -n N – Sets the new niceness value; -p PID – Specifies the process.

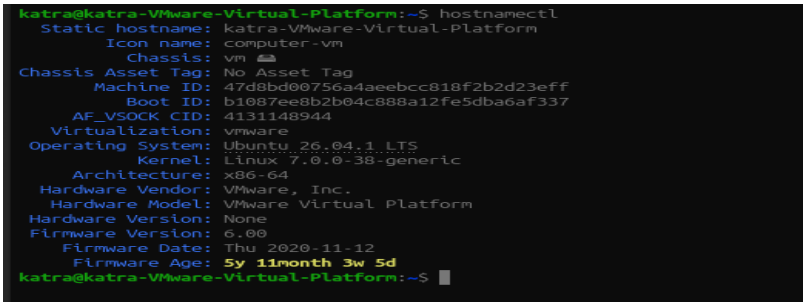


3. SYSTEM INFORMATION

20. uname – Displays kernel and system information. -r – Kernel release; -m – Machine architecture; -n – Hostname; -a – All available system information.

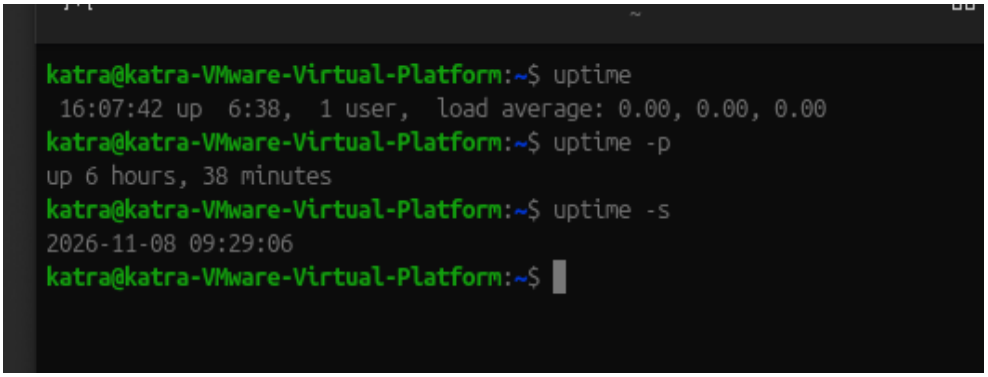


21. hostnamectl – Displays or manages system hostname information. hostname – Displays the hostname.

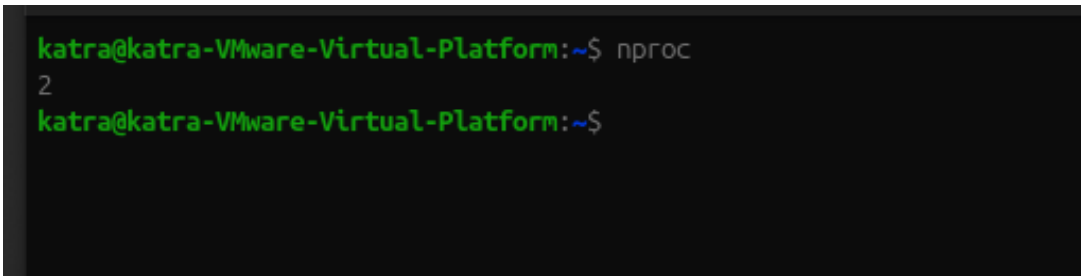


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22. uptime – Displays current time, system uptime, logged-in users, and load average. -p – Human-readable uptime; -s – System boot time.



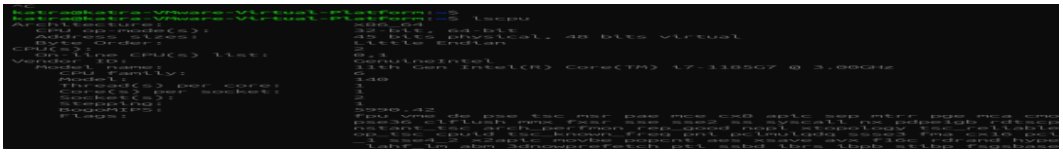
23. nproc – Displays the number of processing units available to the current process.



24. free – Displays RAM and swap memory usage. -h – Human-readable units; -m – Displays values in MiB; -s N – Repeats the display every N seconds.

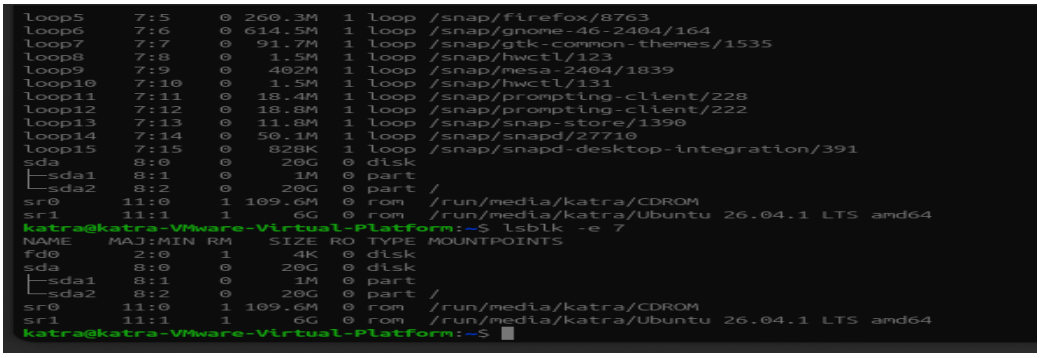


25. lscpu – Displays detailed CPU information, including architecture, model, and CPU count.



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26. lsblk – Lists block devices such as disks and partitions. -e 7 – Excludes devices with major device number 7; -d – Displays whole disks only; -o COLS – Selects displayed columns.



27. ls – Lists files and directories. -l – Displays detailed file information.

28. cat – Displays the contents of a file.

29. head – Displays the beginning of a file. -n N – Displays the first N lines.

30. grep – Searches for matching text or patterns. -m N – Stops after N matching lines; -c – Counts matching lines.

31. less – Displays file contents one page at a time. Space – Next page; / – Searches for text; q – Quits.

32. /proc – A virtual filesystem containing process and kernel information. cpuinfo – CPU information; meminfo – Memory information; uptime – System uptime; loadavg – System load averages; PID/status – Process status and identifiers; PID/comm – Process name; PID/cmdline – Process command line; PID/exe – Link to the process executable.

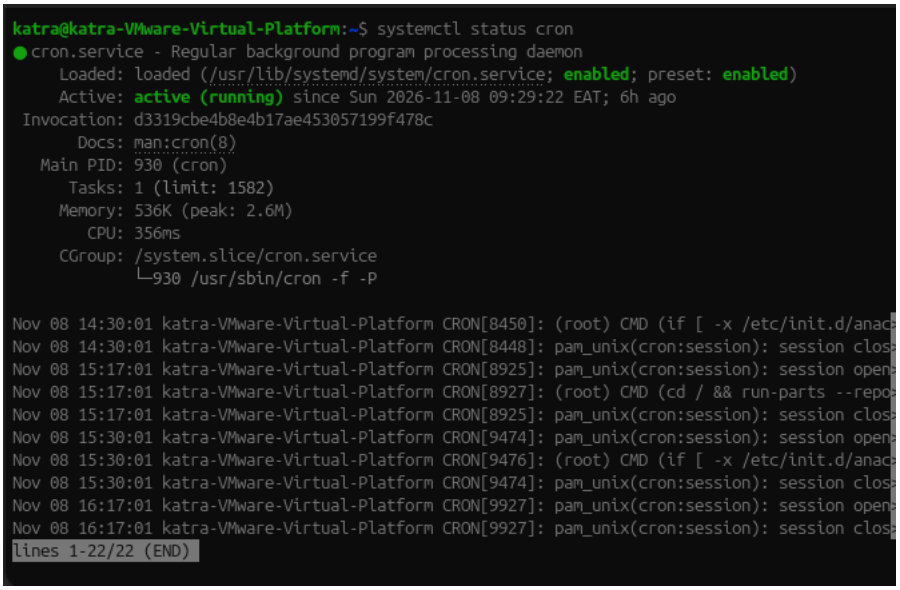
33. /sys – A virtual filesystem providing hardware, device, driver, and kernel information. block – Block devices; class – Device classes; class/net – Network interfaces; bus – System buses; devices – Device hierarchy; class/net/INTERFACE/address – Network interface MAC address.

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4. **SERVICES WITH SYSTEMD**

34. systemd – The Linux system and service manager. It runs as PID 1 and manages system services.

35. systemctl – Controls and manages systemd services. status – Displays service status; is-active – Checks whether a service is running; is-enabled – Checks whether a service starts automatically at boot; start – Starts a service; stop – Stops a service; restart – Restarts a service; reload – Reloads service configuration without stopping it, when supported; reload-or-restart – Reloads if supported or restarts otherwise; enable – Enables automatic startup at boot; disable – Disables automatic startup at boot; --now – Also starts or stops the service immediately; list-units – Lists loaded units; --type=service – Displays service units; --state=running – Displays running services; --state=failed – Displays failed services; cat – Displays a service unit file; show – Displays service properties; -p MainPID – Displays the main process ID.



36. sudo – Runs a command with administrative privileges.

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5. LOGS, BOOT, AND TROUBLESHOOTING

37. journalctl – Displays systemd journal logs. -n N – Displays the last N entries; -b – Displays current boot logs; -p err – Displays errors and more severe messages; -u UNIT – Displays logs for a specific service; --since – Displays logs from a specified time; -e – Jumps to the end; -x – Adds explanatory information; -f – Follows new log entries; -k – Displays kernel messages only; _COMM=sudo – Filters entries associated with the sudo command.



38. logger – Writes messages to the system log or journal.

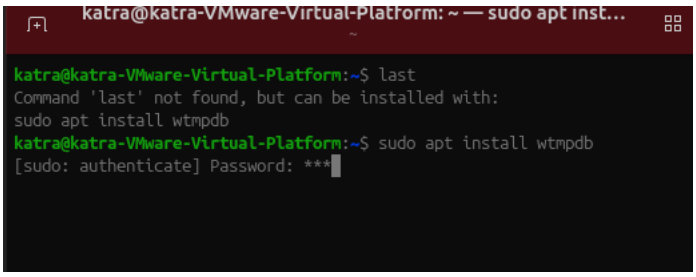


39. /var/log – Directory that stores system and application log files. syslog – General system messages; auth.log – Authentication and login records; kern.log – Kernel messages; journal/ – Systemd journal files.

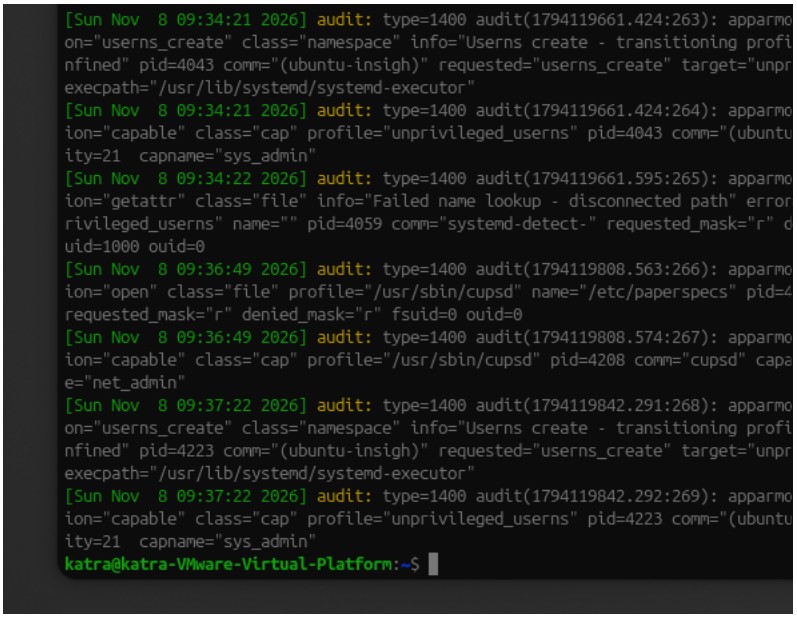
40. tail – Displays the end of a file. -n N – Displays the last N lines; -f – Follows new lines as they are added.

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41. last – Displays previous user login and system reboot records. -n N – Displays the last N records.

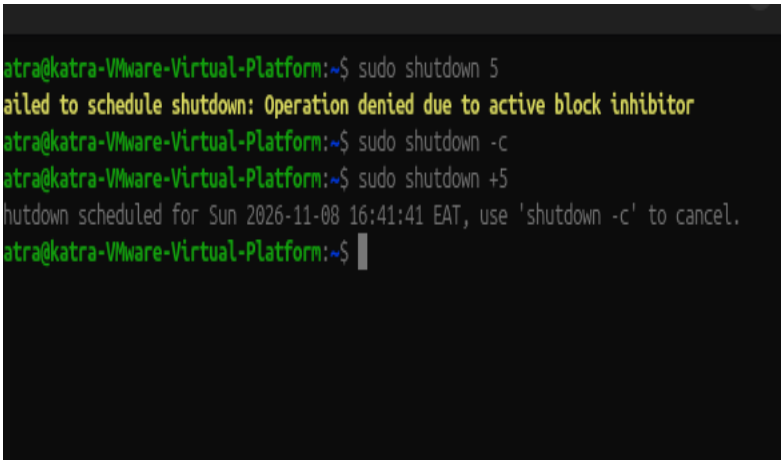


42. dmesg – Displays Linux kernel messages. -T – Displays readable timestamps.

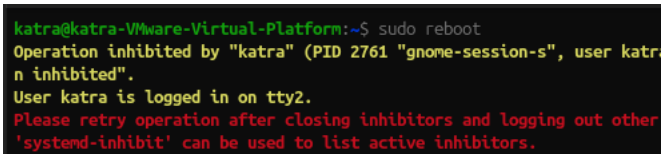


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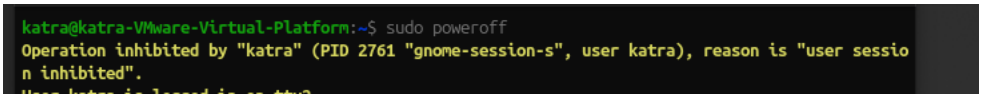
43. shutdown – Shuts down or restarts the Linux system. now – Shuts down immediately; +N – Schedules shutdown after N minutes; -r – Restarts instead of powering off; -c – Cancels a scheduled shutdown.



44. reboot – Restarts the Linux system.

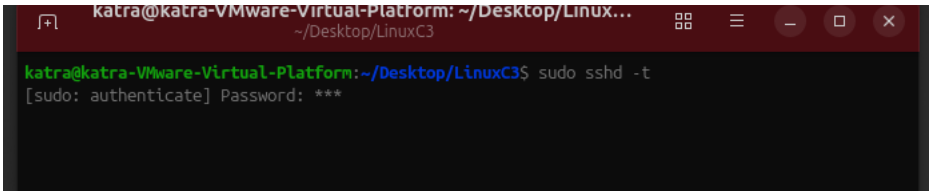


45. poweroff – Shuts down and powers off the system.



46. command -v – Checks whether a command is available and displays its executable path.

47. sshd -t – Tests the SSH server configuration. -t – Checks configuration for errors without starting the SSH service.



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IMPORTANT TERMS

PID (Process ID): A unique number assigned to a process.

PPID (Parent Process ID): The PID of the process that started another process.

SIGTERM: A signal that requests a process to terminate.

SIGKILL: A signal that forcefully terminates a process.

SIGINT: A signal used to interrupt a process.

SIGTSTP: A signal used to pause a foreground process.

Niceness: A value that influences CPU scheduling priority.

Service: A background program that provides a system function.

systemd: The system and service manager that starts and manages services.

Journal: The logging system used by systemd.

Load Average: The average number of processes running or waiting for CPU resources over 1, 5, and 15 minutes.

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